



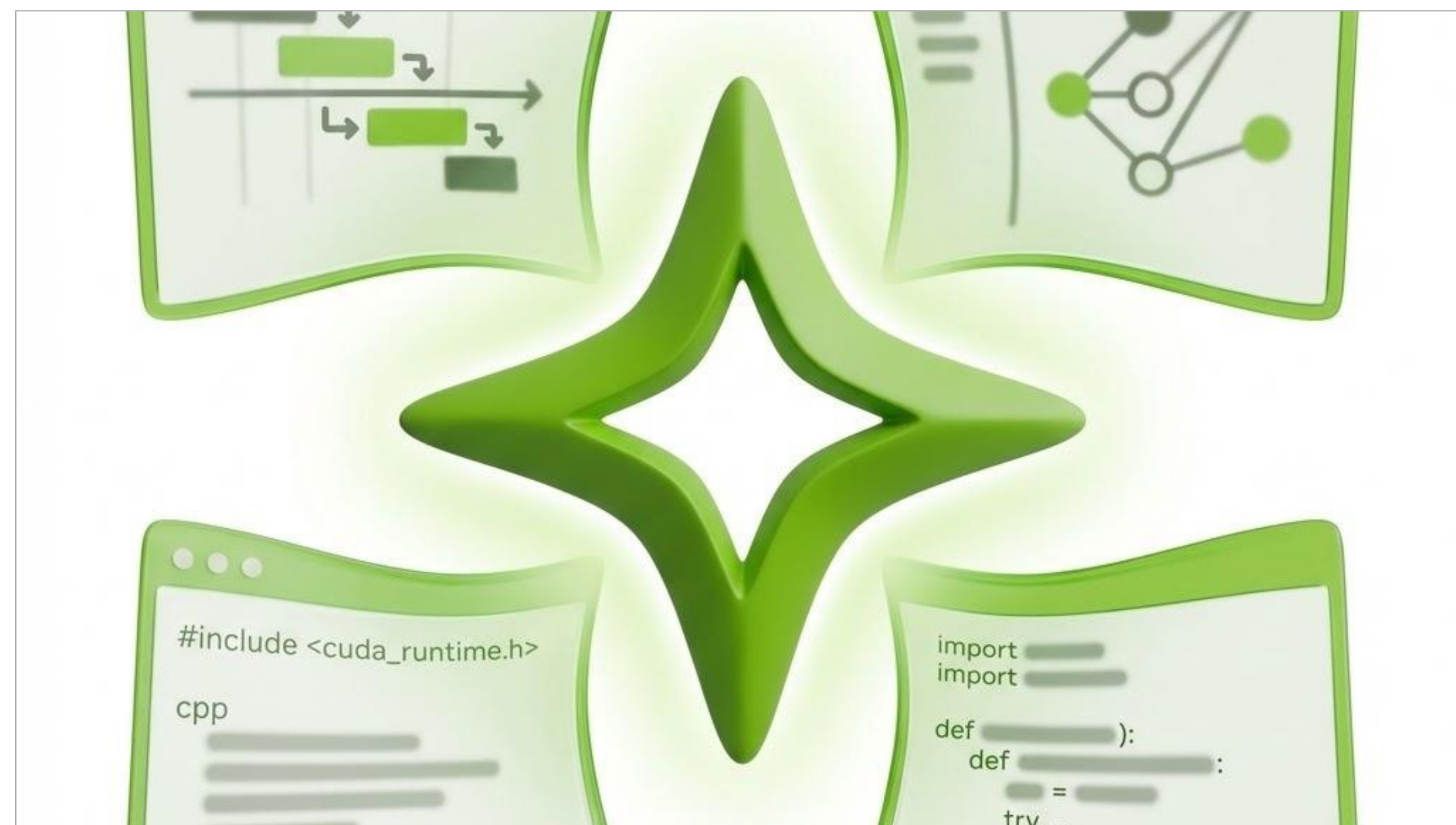
AI Coding for the GPU: **Building a Coding Agent to Help GPU developers write Speed of Light Code**

Matt Frazier, Engineering Director – AI Developer Tools
GTC 26 | March 19, 2026



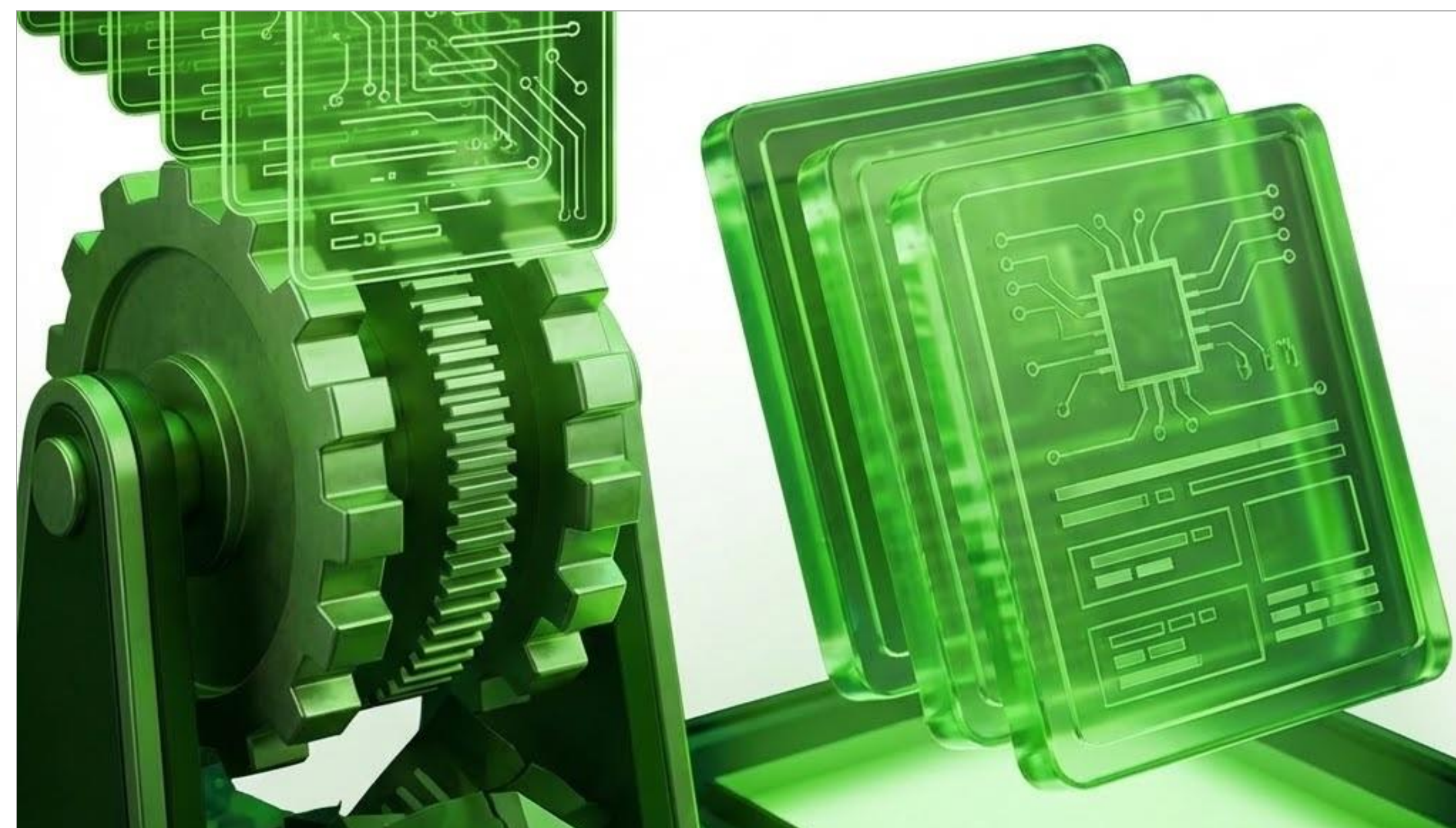
AI Developer Tools

Vision



Deliver the CUDA Intelligence Ecosystem

Nsight Tools, IDE, Compute, Systems Key Evaluation Services and FOSS Blueprints & Reference Implementations



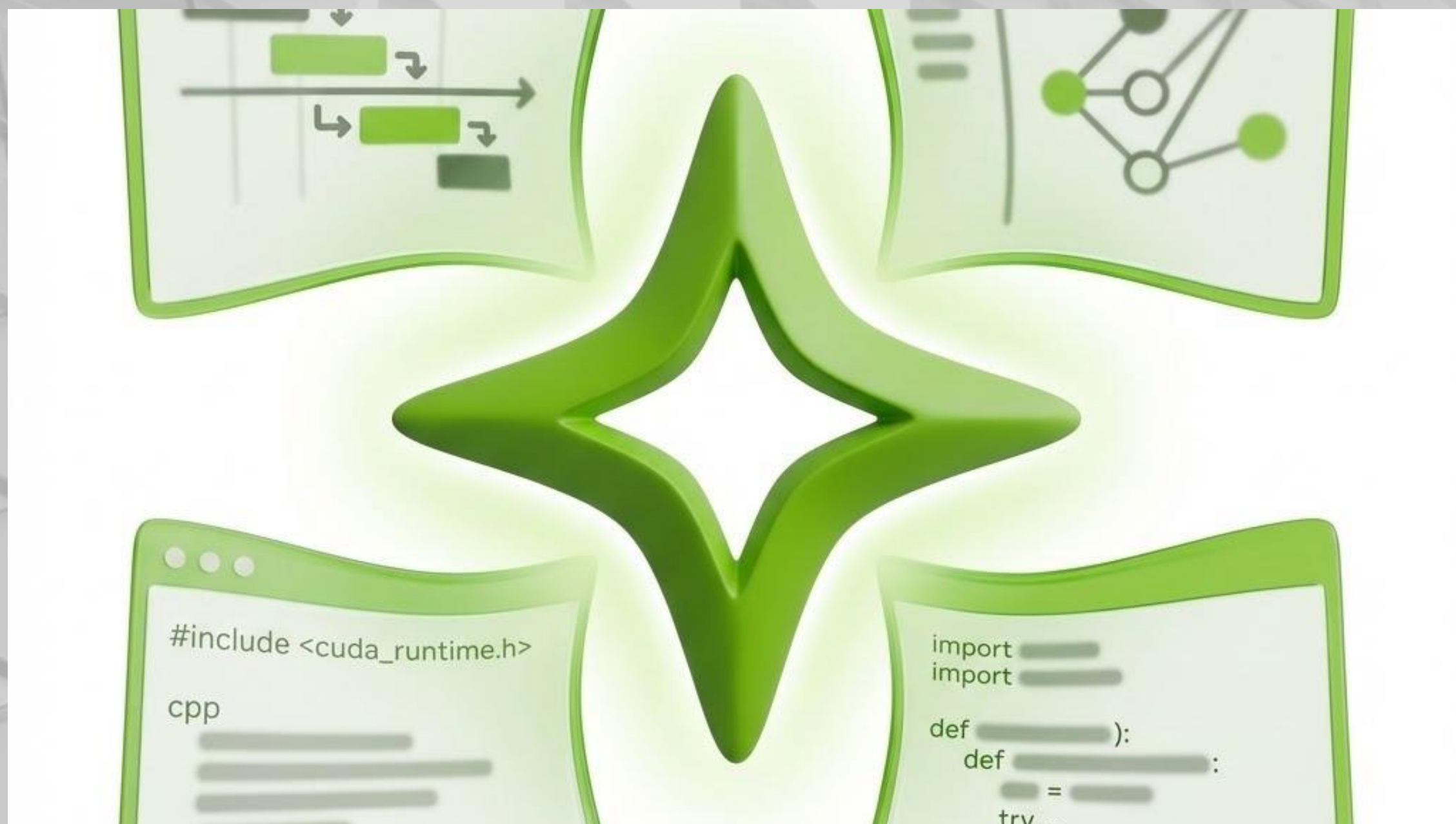
Filter Noise and Verify with Evals

Extracts signal from noise, which CUDA Intelligence is Applicable Expertise in your Domain



Power Professional Agentic CUDA Engineering

Machines test, propose, evaluate, select and evolve



Deliver the CUDA Intelligence Ecosystem

Net New Knowledge

Deliver the CUDA Intelligence Ecosystem

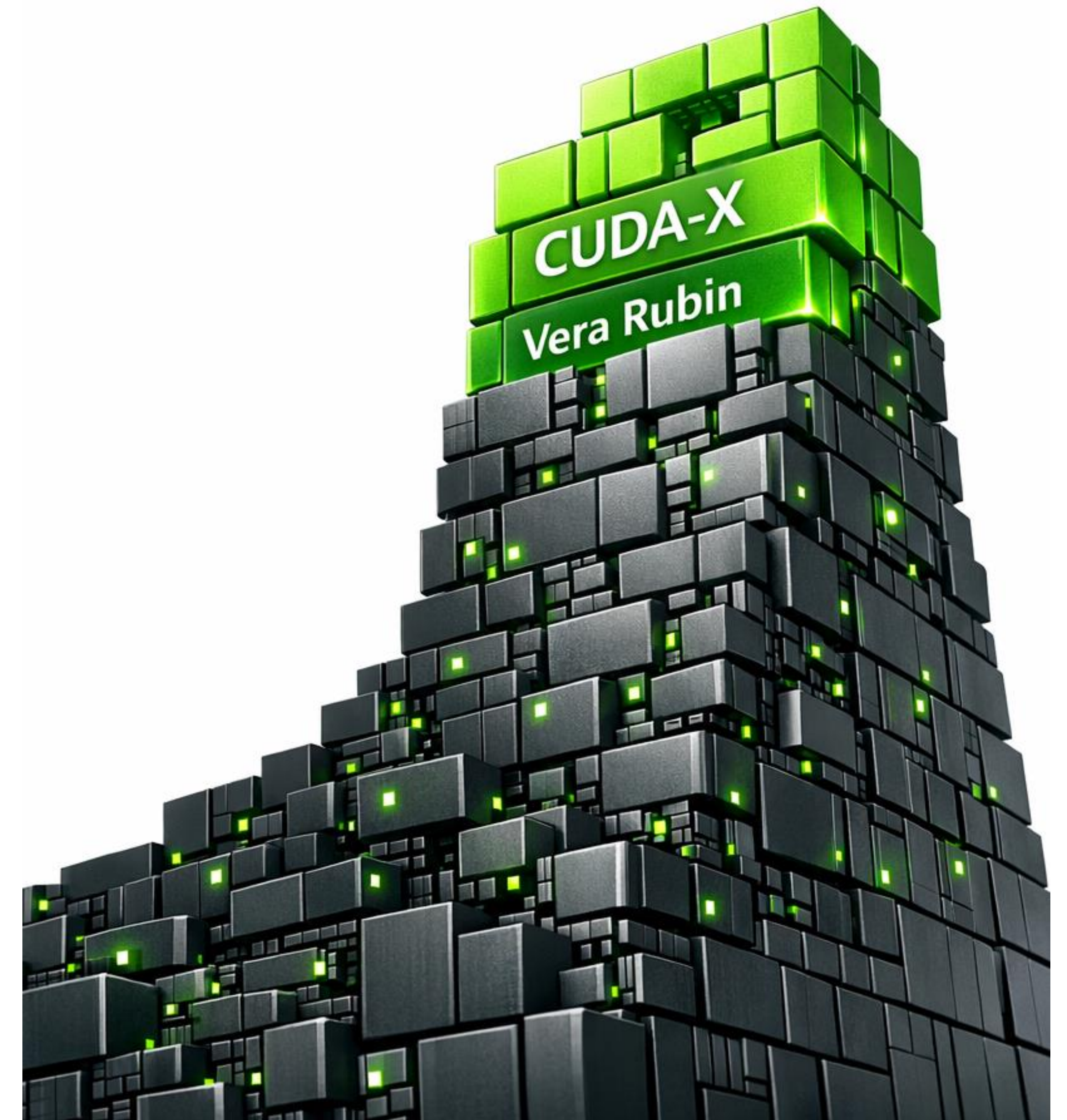
Annual, often faster

Measuring NNK: A Test-Driven Development (TDD) approach to evals.

Setting a new baseline expectation for CUDA developers

Get out of our own way and yours

**“70+ CUDA-X
Libraries this year”
plus Vera Rubin -
JHH**



Meet you where you are

Deliver the CUDA Intelligence Ecosystem

Power where you're going with open tools

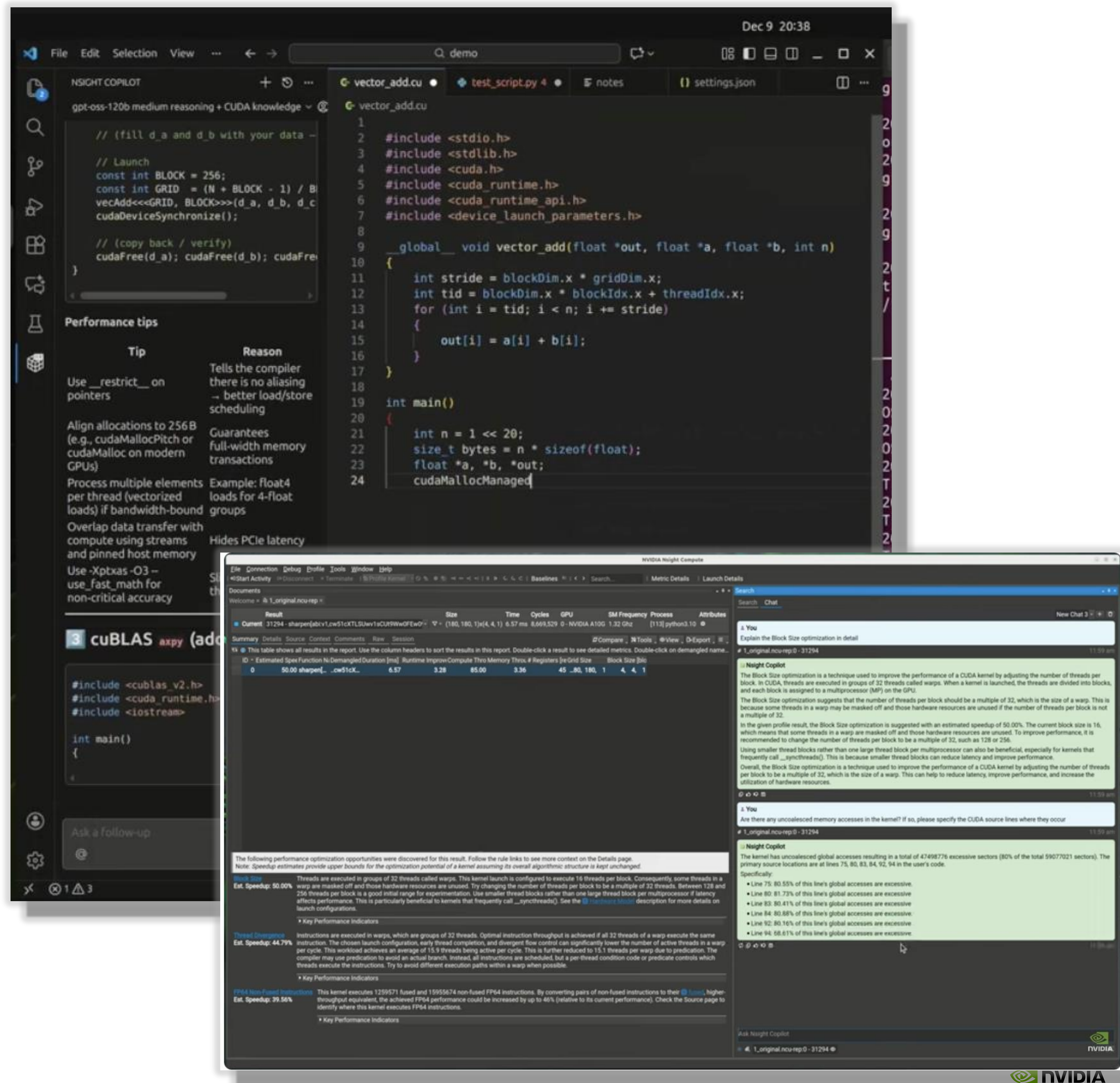
Where you are - the IDE

- VS Code, Cursor

Where you're going - Agentic Coding

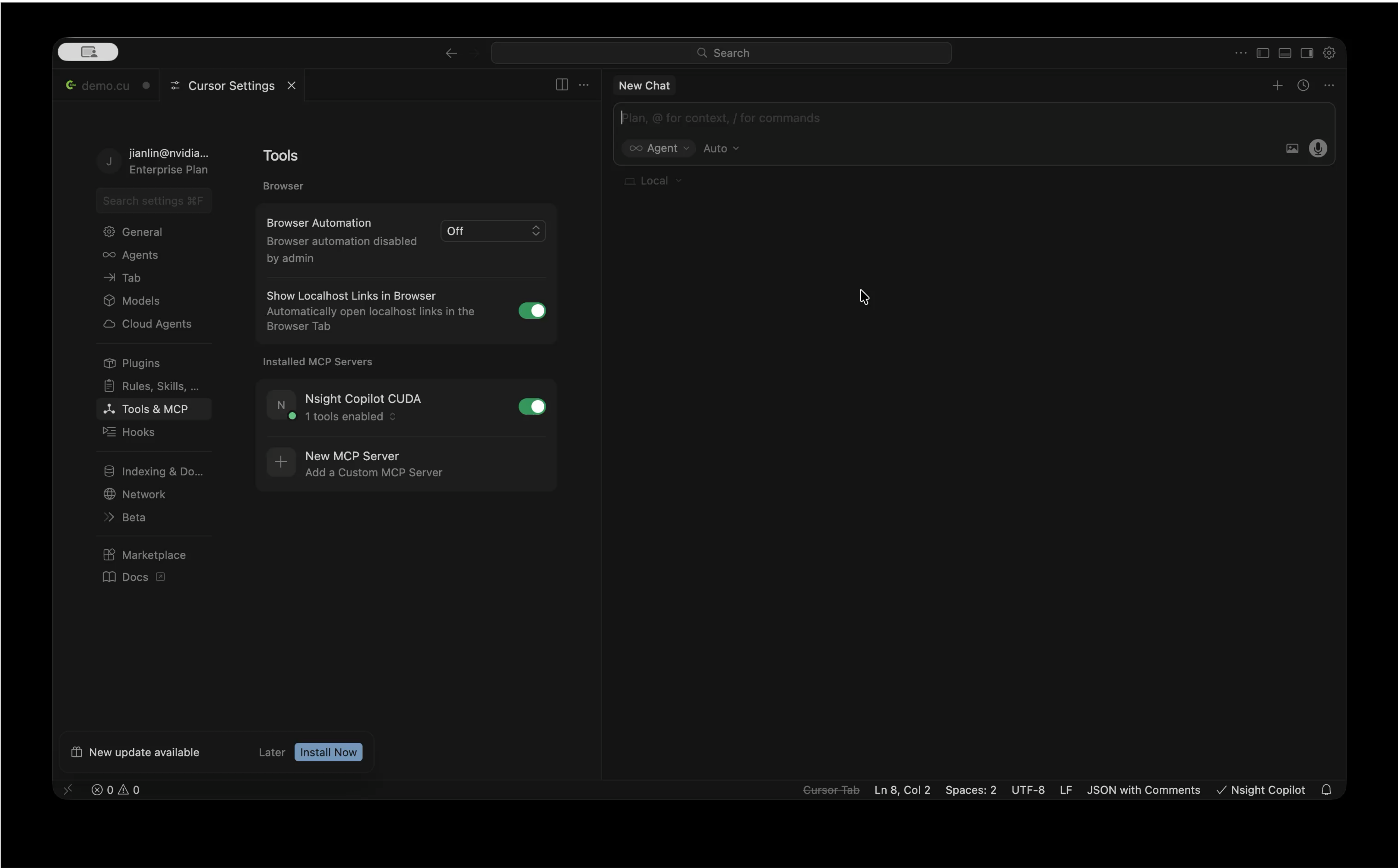
- Claude Code, codex, Gemini CLI, opencode
- Nsight Copilot inside Nsight Compute
- Guiding investigations, zeroing in on profiling targets, and eventually optimizations.

Demo: Claude Code with Nsight CUDA Intelligence



Nsight CUDA Intelligence MCP in Cursor

Deliver the CUDA Intelligence Ecosystem



Deliver Intelligence + Measure Performance

Deliver the CUDA Intelligence Ecosystem

We commit to

- Build the reference implementation & get out of your way
- Productize what needs NVIDIA scale & expertise

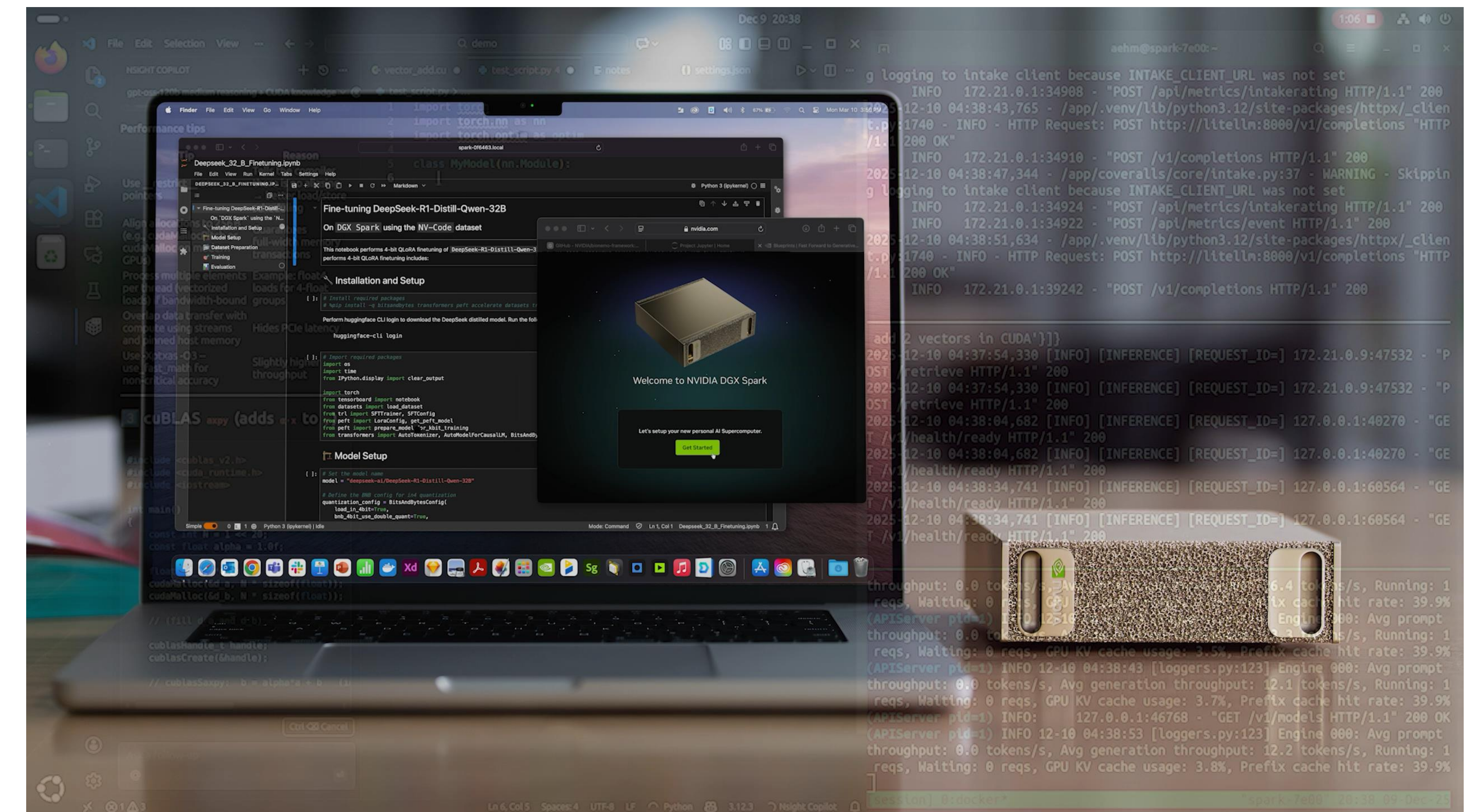
Delivering Intelligence

- Full stack FOSS AI v RAG
- -> MCP (Model Context Protocol)
- -> SKILLs
- -> Agents select Skillsets, intrinsic evals

Agentic Coding - Performance RL

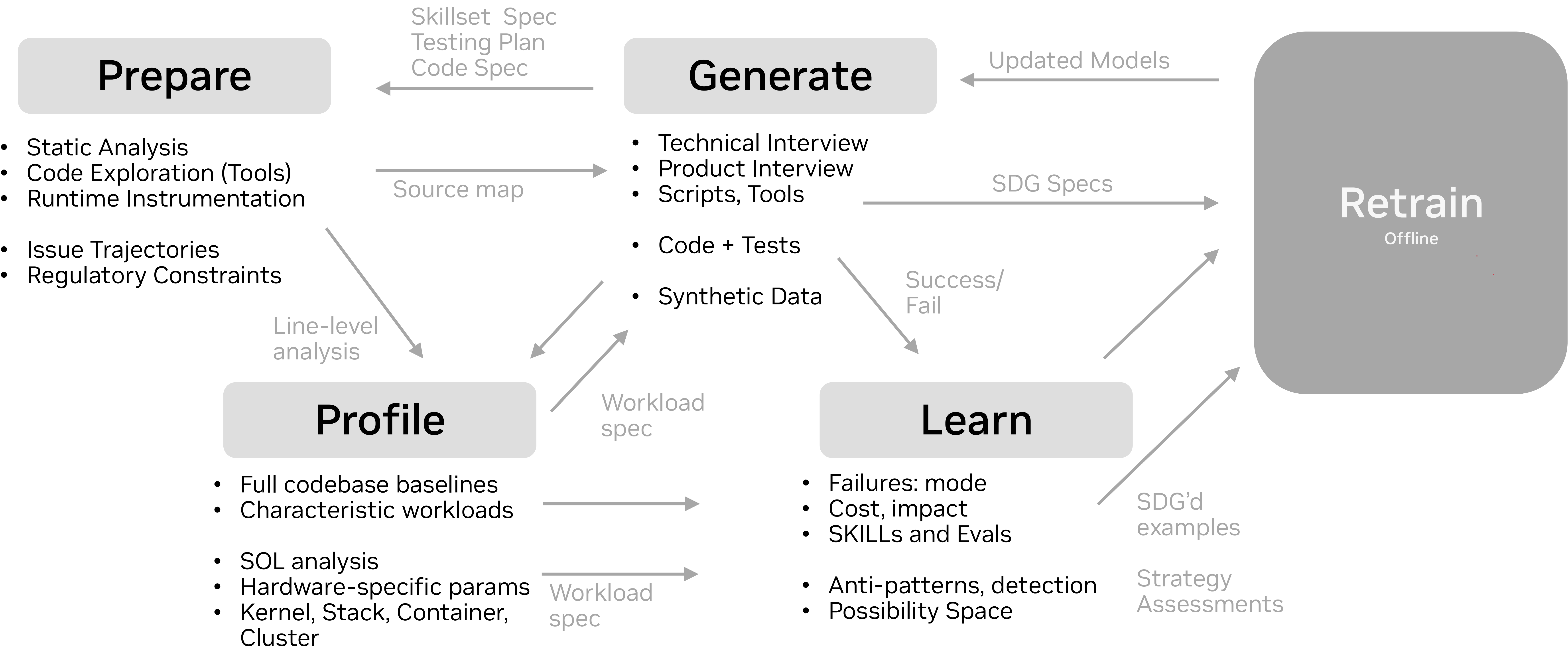
- Balance Speed & Memory Pressure
- Also token cost, time
- Maintainability - CUDA-X, PTX Fragile

Blueprints and Open Horizontals



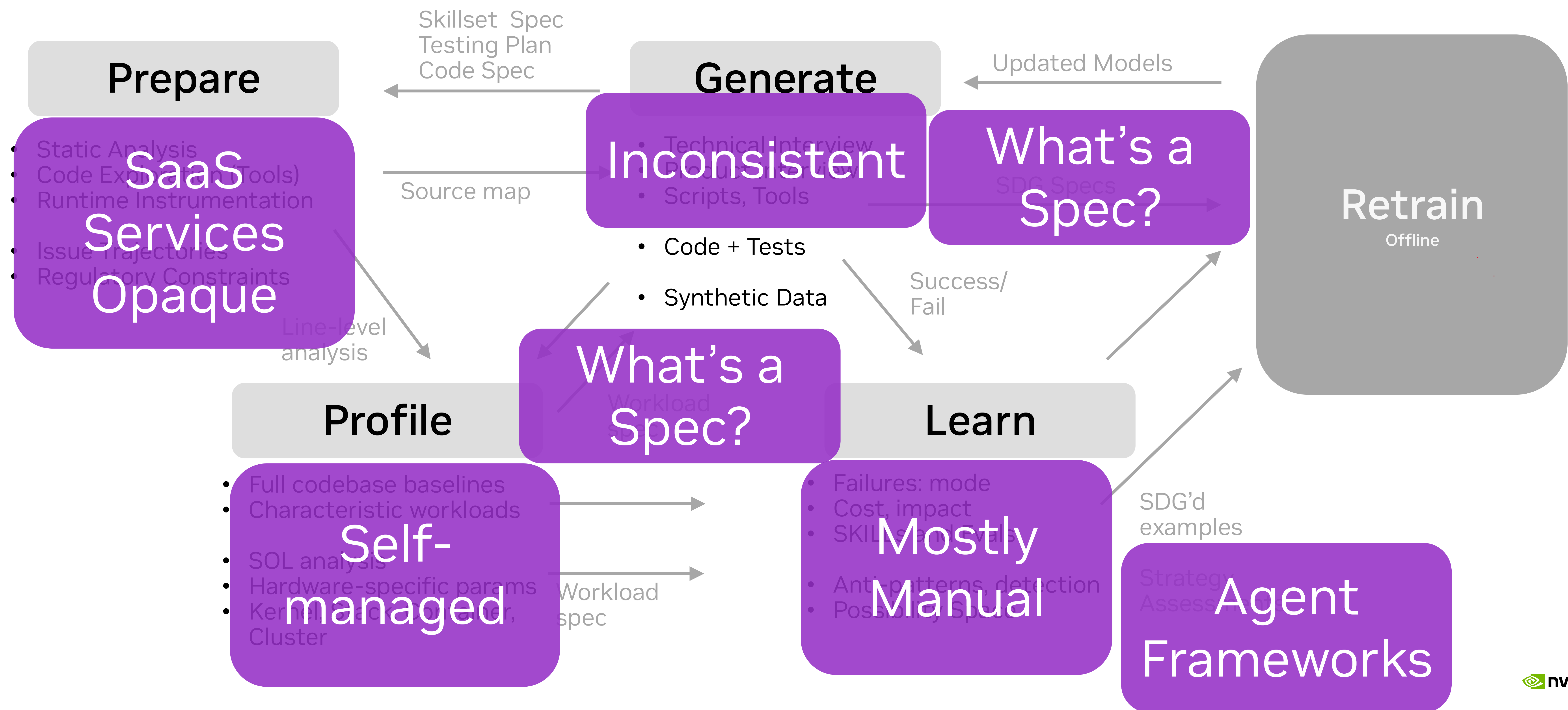
The CUDA Intelligence Ecosystem

Deliver the CUDA Intelligence Ecosystem



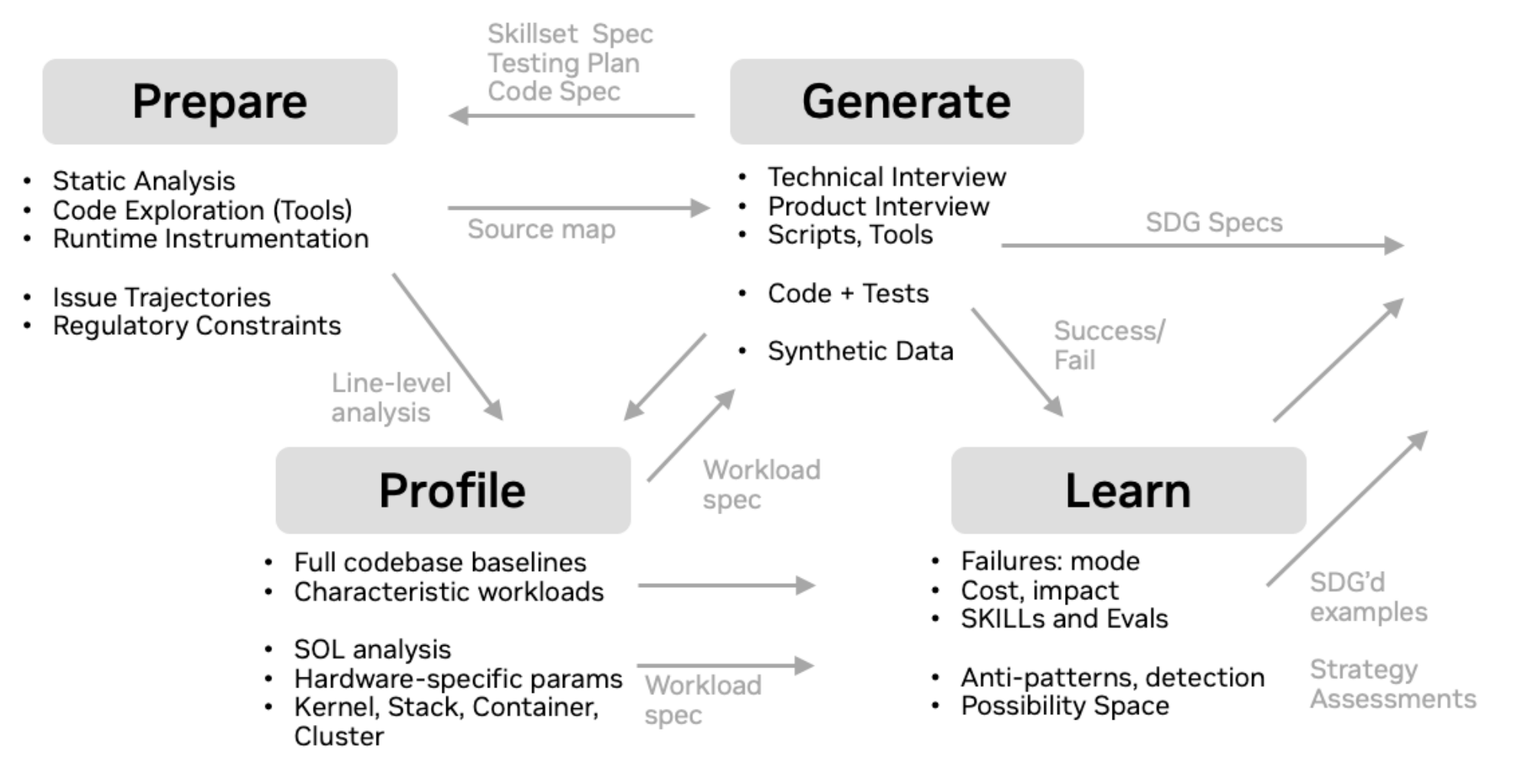
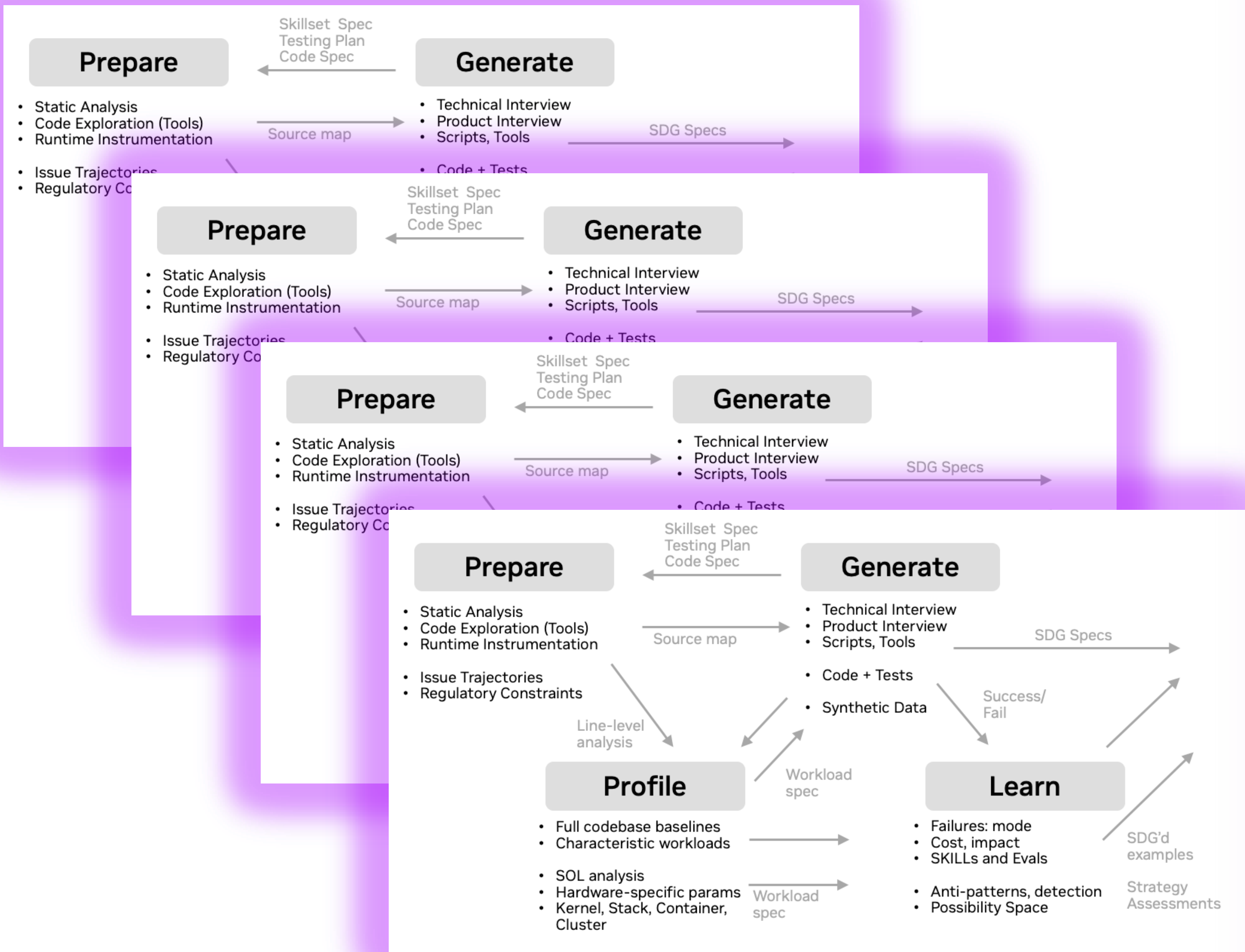
The CUDA Intelligence Ecosystem - Today

Deliver the CUDA Intelligence Ecosystem



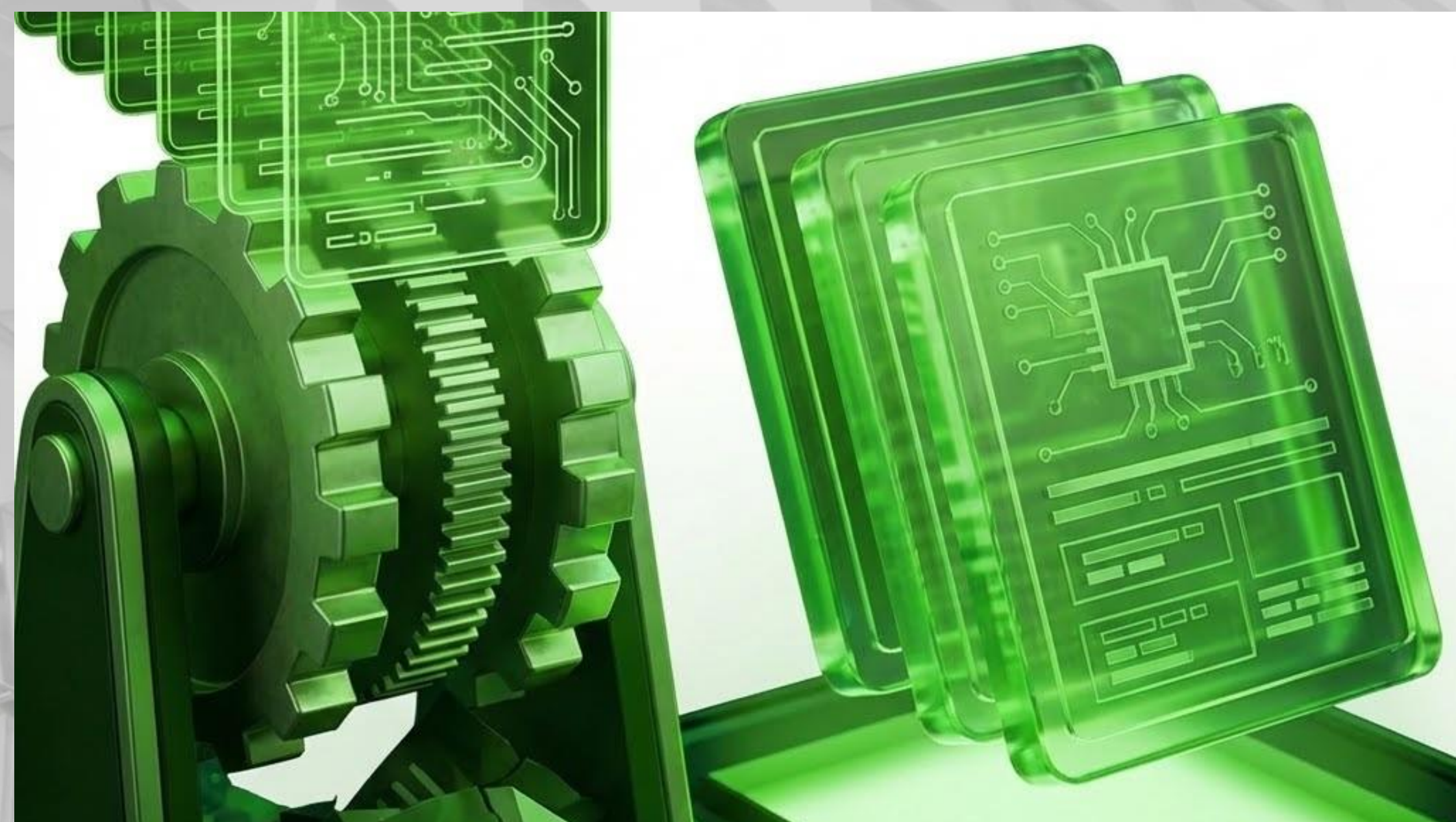
The CUDA Intelligence Ecosystem

Deliver the CUDA Intelligence Ecosystem



Learn

Retrain
Offline



Filter Noise and Verify with Evals

Evals Separate Signal from Noise

Filter Noise and Verify with Evals

Generate → **Profile** → **Optimize** → **Retrain**



Self-reflective agents: Models debug and revise their own kernel output.

Loop automation: More of the improvement loop runs without manual intervention.

2026 net effect: The flywheel spins faster, but there is still substantial performance headroom.

computeeval.nvidia.com/leaderboard Live Preview (Mock)

Architecture: B200 + H100 CUDA: 13.1 Domain: AI Kernels Sort: Overall Score Open-weight only: Off [Apply Filters](#)

#	Model	Arch	CUDA	Domain	Correctness	Overall
1	Claude Opus 4.6 (thinking)	B200	13.1	AI kernels	93.0%	91.2
2	Gemini 3.1 Pro	H100	13.1	AI kernels	86.8%	87.4
3	GPT 5.2 high	H100	13.1	AI kernels	83.1%	84.6
4	KernelBot-SFT-v2 (open)	B200	13.1	AI kernels	76.4%	79.1
5	gpt-oss-120b-high	H100	13.1	AI kernels	68.5%	73.2

Data source: ComputeEval + KernelBot (illustrative mock)

Last updated: Jan 2026

Architecture
Filter by GPU family and compute capability

CUDA Version
Target specific toolkit versions and features

Problem Domain
AI kernels, graphics, scientific computing, and more

Scenario Match

Architecture
B200, H100

CUDA Toolkit
13.1 (target)

Recommended model
Claude Opus 4.6
Best correctness + strong performance under selected filters

ComputeEval

- Accelerated Computing is for Performance
- CUDA-wide Correctness & Performance Evals
- Generating Characteristic Workloads
- Use libraries balance maintainability

ComputeEval
NVIDIA



500+ handcrafted problems

Breadth and correctness across the full NVIDIA software ecosystem

AI kernels · Graphics · CUDA-X · Math libraries

All of CUDA vs Targeted

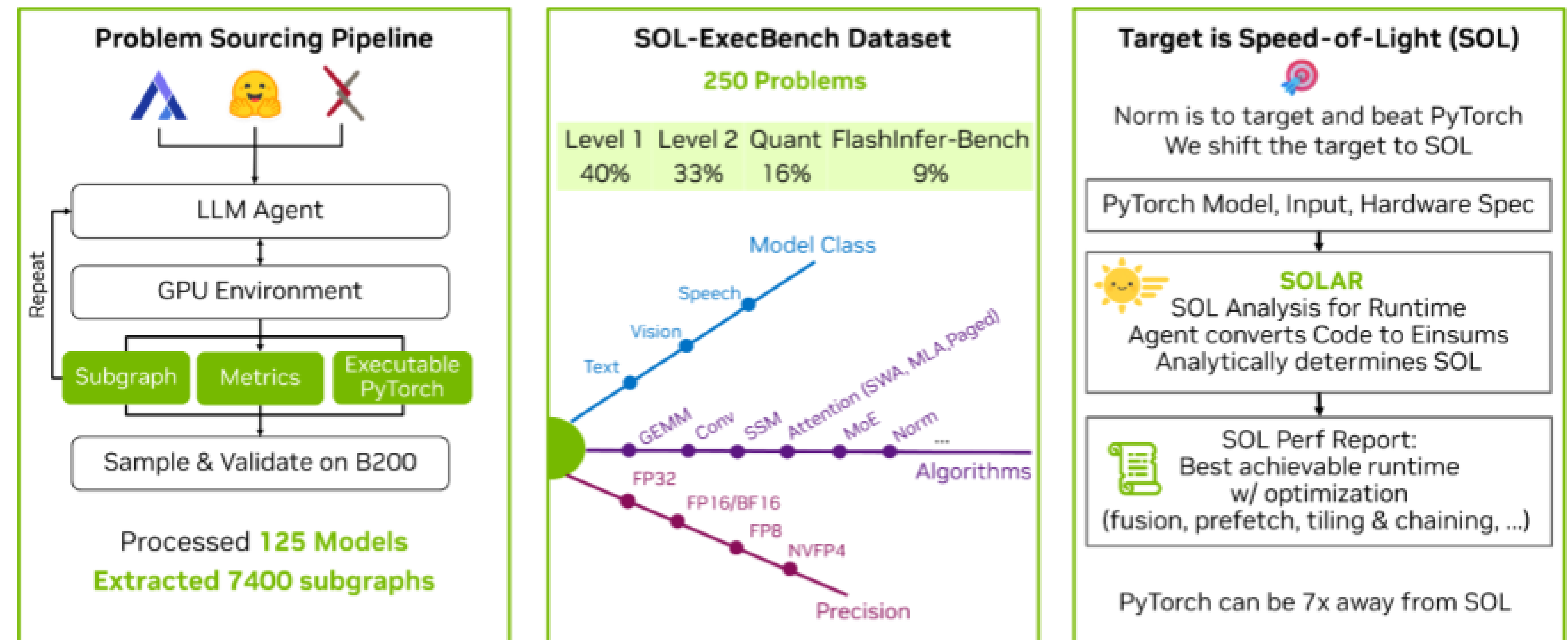
Filter Noise and Verify with Evals

ComputeEval - horizontal, all of CUDA

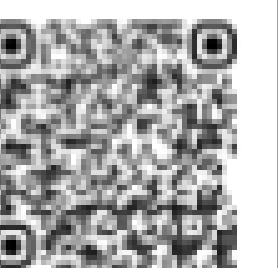
Research - vertical, targeted

- SOLExecBench from ATLAS
- Targeted Kernels vs SOL
- One of several efforts

SOL-ExecBench: Speed-of-Light Benchmarking for Real-World GPU Kernels Against Hardware Limits



<http://research.nvidia.com/benchmarks/sol-execbench>



The Elephant in the Room

Filter Noise and Verify with Evals

Performance Isn't Optional

Profiling as a Service

- Test-time RL
- Train time RL

Wall Clock is a proxy

- Memory pressure, occupancy
- Tokens/watt, tokens/MIP

Systems vs Compute - Your Domain

- Hardware targets
- Characterized Workloads





Power Professional Agentic CUDA Engineering

The Era of Automatic Bespoke Software

Power Professional Agentic CUDA Engineering

Your Problem Domain

Domain & Skill Evals

- Frontmatter, provenance, security
- Backmatter, breadth, tools

Enterprise Expertise <-> Domain Expertise

The era of bespoke software

SKILLs bring knowledge, **you add expertise**

Recapitulating Computer Science

- KV-Caches, Memory Boundaries, Compaction
- The why: **Your problem domain sets requirements**

Evolving SKILLs interactively

- Natural compaction boundary

SKILLs and TRACEs

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SKILL structure

- Brief, progressively discoverable

Real World

- System prompt is still finite
- Skill discovery still probabilistic
- Best practice: maximize determinism

TRACE.md

Summary of a trajectory from inception through runtime, exercise, and resolution.

- Product of a development trajectory + acceptance evaluation
- Iteratively discoverable
- Agent-friendly, human compatible
- Capture successes, failures, verification tools, strategies

```
skill-name/  
├── SKILL.md           # Required: metadata + instructions  
├── scripts/          # Optional: executable code  
├── references/        # Optional: documentation  
├── assets/           # Optional: templates, resources  
└── ...               # Any additional files or directories
```

```
trace-context-ts/    # bot-kernel-profile-iter-2021509  
├── TRACE.md         # Required: metadata + results  
├── scripts/         # Optional: executable code  
├── logs/            # Optional: tool trace logs, outputs  
├── assets/          # Optional: test data, workloads  
└── ...              # Any additional files or directories
```

What's Coming

Power Professional Agentic CUDA Engineering

Probabilistic agents for resilience

- “Compile” into Deterministic tools & data

"Learn" from failures & successes

- SKILLs -> Agents -> TRACEs -> Learning/Strategy
- "Compile" into Deterministic
- Left-shift specifications
- And resist temptation

/results/fsh-h1100-opt-002/TRACE.md

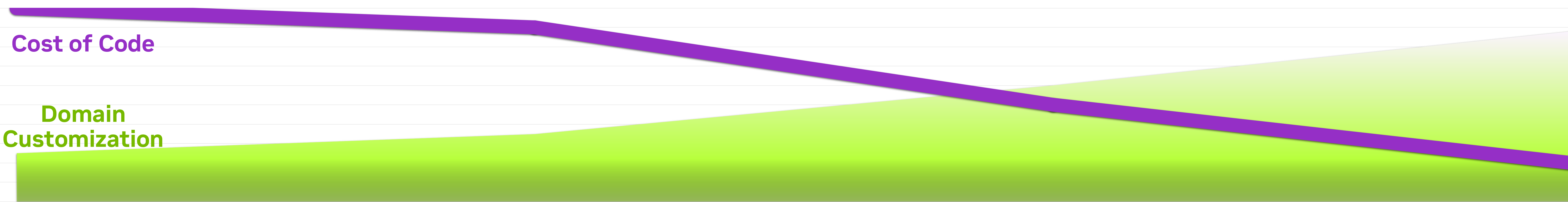
```
---
id: fa4-h100-opt-082
objective: Outperform FlashAttention-4 throughput on H100 (Hopper)
status: partial
skills-used: cuda-expert nv-nsight-ops kernel-tiling-strategy
metrics:
  target: 950 TFLOPS
  achieved: 892 TFLOPS
  occupancy: 82%
resources:
  tokens: 142,500
  wall_clock: 02h:15m:10s
  compute_footprint:
    cpu_time: 18m:23s      # Intensive template metaprogramming & JIT compilation
    gpu_time: 01m:10s     # Actual kernel execution across Nsight profile passes
    memory_peak: 4.2GB
metadata:
  compiler: nvcc-12.4
  arch: sm_90a
  toolkit: ncu-2024.1
---

# Execution Trajectory

### 1. Executive Summary
The attempt reached **840 TFLOPS**, falling short of the FA4 benchmark (~900+ TFLOPS).
While compute utilization (SM SOL) was 94%, the bottleneck was identified as **register
spills** during the softmax scaling phase.
```

CUDA Agentic Engineering

Power Professional Agentic CUDA Engineering



2024	2025	2026	2027	2028
LLMs Generate Code	Nsight CoPilot VS Code CUDA Intelligence	Think in Codebases NCU+MCP+SKILLs	Agents Create & Consume Flywheels	Bespoke Automatic Development
	NSight Copilot VSCode	NSight Copilot VSCode, Cursor, Claude MCP, SKILLs, TRACEs	NSight Copilot +NSight Systems Codebase Transform	NSight Copilot Throughout your SDLC
	"CUDA Intelligence" RAG ComputeEval	"CUDA Intelligence" Agentic ComputeEval	"CUDA Intelligence" Net New Knowledge TRACE to STRATEGY	"CUDA Intelligence" Net New Knowledge in every Mode
		Blueprints Your Infra, DGX Spark Cloud Profiling Reference	NSight Cloud Profiling NVIDIA Hardware Breadth	

The Long Tail of Innovation

Power Professional Agentic CUDA Engineering

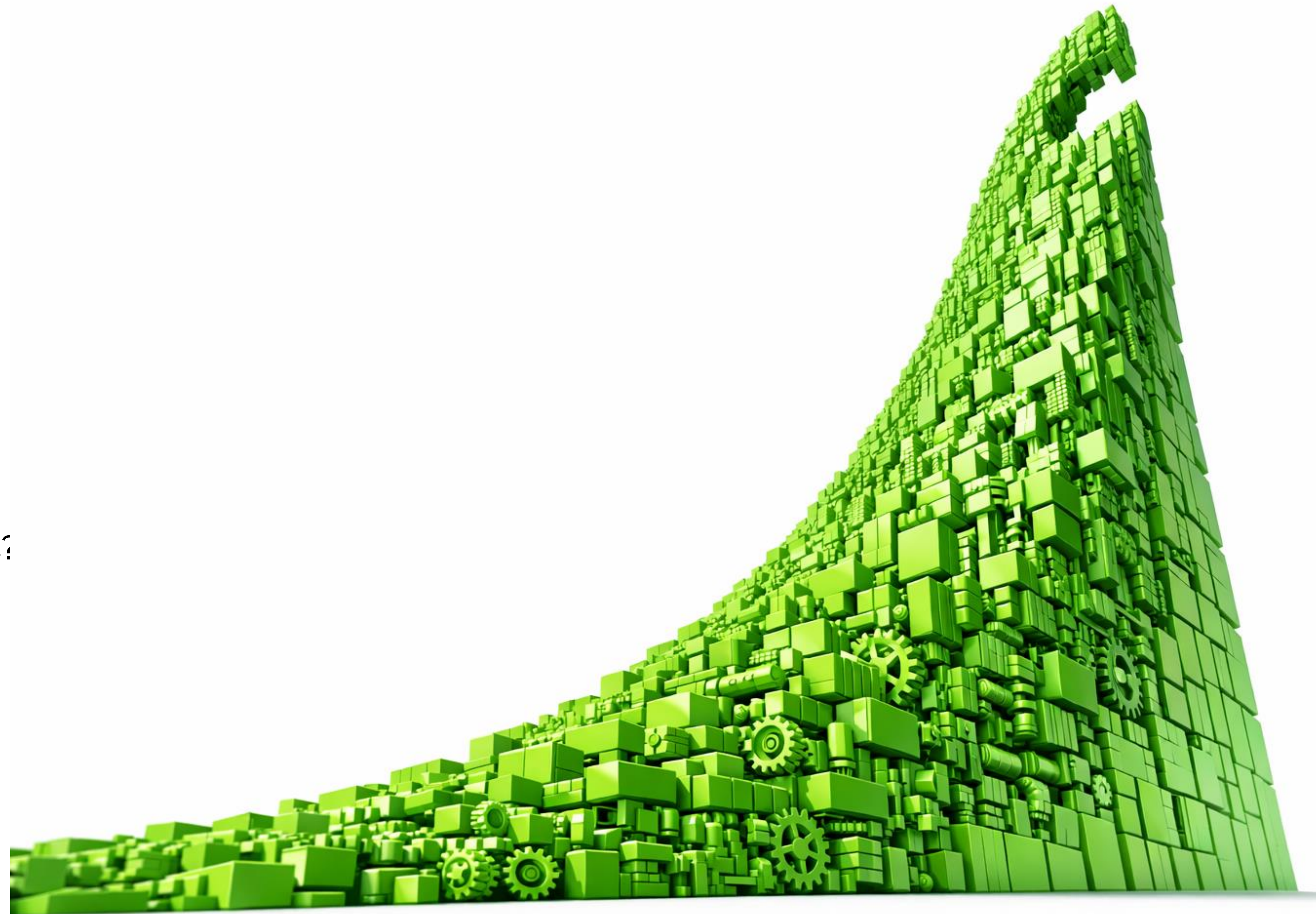
NVIDIA flexible enough for What's Next

- Next transformer, diffusion
- Whatever Yann is doing

The long-tail of Innovation - the next F5

So much in the last year

- Last year: Are we trying to replace CUDA Engineers?
- Accountants with Calculators
- Engineers with Agents



Developer Tools Across GTC

Sessions

- [S81590](#): Lowering the Barriers: How New Tools Unlock GPU Acceleration for Everyone
- [S81831](#): AI Coding for the GPU: Building a Coding Agent to Help GPU developers write Speed of Light Code
- [S81653](#): LLM-Generated CUDA Kernels: Are We There Yet?
- [S81772](#): Don't leave Tensors on the Table: Programming and Optimizing Tensor Cores

Labs

- [DLIT81545](#): Profiling Python and AI workloads with Nsight Compute
- [DLIT81641](#): Find the Bottleneck – Optimize AI pipelines with Nsight Systems
- [DLIT81579](#): Optimizing PyTorch models for high-performance inference with Nsight Deep Learning Designer

Connect with the Experts

- [CWES81549](#): What's in Your Developer Toolbox? CUDA, AI, and Graphics Profiling, Optimization, and Debugging Tools
- [CWES81535](#): CUDA Developer Best Practices
- [CWES81771](#): How to Run and Optimize Your Workloads on the NVIDIA Grace and Vera CPUs
- [QA82372](#): What's in Your Developer Toolbox? CUDA, AI, and HPC, Optimization, and Debugging Tools

Live Demos

Come and visit the Developer Tools pod at the NVIDIA booth during exhibition floor hours!

Developer Tools are **free** and packaged in the latest version of the CUDA Toolkit

<https://developer.nvidia.com/cuda-downloads>

Support is available via:

<https://forums.developer.nvidia.com/c/developer-tools>

More information at:

<https://developer.nvidia.com/tools-overview>

